Data Quality

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Types of business data

- OLTP, relational
- Data warehouses, star, hierarchical/multidimensional
- Semi structured, e.g. json
- Unstructured, e.g. emails
- Master data
- Metadata

Data quality dimensions

- Hard dimensions
  - Completeness—do we have all the data?
  - Accuracy—are the values accurate?
  - Consistency—do data contradict among different systems?
- Soft dimensions
  - From the users' perspective, e.g. trust.

Understanding data: Data profiling

- Column value distributions and other statistics
- Measure null ratio and length distribution for string types
- Enforced vs. *de facto* functional dependencies
  - Even if a dependency is not enforced by the domain modell, it may be required *de facto*
  - Violating hidden dependencies may lead to **redundancy and inconsistency**

Data profiling in the SSIS Control flow

Profiles (Table View)

- Data Sources
  - .\s2019
    - Databases
      - AdventureWorksDW2016CTP3
        - Tables
          - [dbo].[DimEmployee]
            - Candidate Key Profiles
            - Column Length Distribution Profiles
            - Column Null Ratio Profiles
            - Column Statistics Profiles
            - Column Value Distribution Profiles
          - [dbo].[DimProduct]
            - Candidate Key Profiles
            - Column Length Distribution Profiles
            - Column Null Ratio Profiles
            - Column Statistics Profiles
            - Column Value Distribution Profiles

Column Null Ratio Profiles - [dbo].[DimProduct] Encrypted Connection 1000 Rows

Column	Null Count	Null Percentage
ArabicDescription	210	34.6535 %
ChineseDescription	210	34.6535 %
Class	276	45.5446 %
Color	0	0.0000 %
DaysToManufacture	0	0.0000 %
DealerPrice	211	34.8185 %
EndDate	406	66.9967 %
EnglishDescription	210	34.6535 %
EnglishProductName	0	0.0000 %
FinishedGoodsFlag	0	0.0000 %
FrenchDescription	210	34.6535 %
FrenchProductName	0	0.0000 %
GermanDescription	210	34.6535 %
HebrewDescription	210	34.6535 %
JapaneseDescription	210	34.6535 %
LargePhoto	0	0.0000 %
ListPrice	211	34.8185 %
ModelName	209	34.4884 %

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SQL Server Data Quality Services

- Databases:
  - DQS_MAIN: logic and Knowledge Bases (metadata)
  - DQS_PROJECTS: user projects
  - DQS_STAGIG_DATA: temp storage
- Project types:
  - Basic data cleansing
  - Identity mapping and de-duplicating

Data Cleansing projects

- Kowledge base: a collection of domains
  - Domain name and base type, e.g. string
  - Reference data: domain values
    Leading value + synonyms
  - Normalization procedure e.g. to_lower()
  - Domain rules to be enforced
    - May use a column pattern detected by Data Profiling
  - Term based relations
- Project configuration
  - Min score for suggestions, default 0.7
  - Min score for auto correction , default 0.8

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Data Cleansing results

- Correct
- Corrected
- Suggested (between the two min scores)
- New
- Invalid (violates the domain rules)

Building a Knowledge Base

- Knowledge discovery with a „gold standard” sample
- Manual domain preparation
  - Adding rules
  - Setting Errors and Synonyms
- Publishing and sharing the Knowledge Base

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Case study: BERT based entity mapping

(separate slides)